

Supplemental Material: Symmetry control of exceptional structure in the massive scalar quasinormal spectrum of doubly rotating five-dimensional Myers–Perry black holes

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S1. SOLVERS AND THEIR VALIDATED DOMAINS

r_- is the inner horizon radius; $z_- = r_-^2$. Extremality is $M - (|a| + |b|)^2$, zero at extremality. The independence rule is that any claim requires agreement between one recurrence solver and one recurrence-free solver; two wrappers around one discretization do not count.

Solver	Formulation	Rec.-free	Validated domain	Known failure
A	Leaver continued fraction, Gaussian-reduced	no	$r_- \leq 0.2287$, ex-	drifts $2-6 \times 10^{-4}$ per depth dou-
B	raw recurrence, Hill determinant, Wynn	no	tremality ≥ 0.294	bling in the quasinormal regime
C	exterior complex-scaled Chebyshev collocation	yes	as A	shares A's recurrence; not inde-
D	multidomain near-horizon spectral	yes	as A, provided $\theta_{\min} < 90^\circ$	pendent of A
HP	full <code>mpmath</code> chain incl. coefficients	no	ordinary modes	no admissible contour as $\text{Re } \Omega \rightarrow 0$
			arbitrary precision	same formulation limit as C
				slow

TABLE SI. Solver inventory. Solver C's independence is enforced structurally: a test greps the module for the recurrence routines and fails if any appears.

The raw Frobenius recurrence has width 13 for generic two-spin parameters and 9 in the static case; the reduction to three terms is exact. The leading large- n tail is $R_n = 1 + u_1 n^{-1/2} + \dots$ with $u_1 = -\sqrt{-2c}$, $c = i\Omega(r_+ - r_-)$, which reduces the required depth by a factor ≈ 1.35 . Higher tail orders are not available: the Gaussian reduction degrades past $n \sim 500-800$, where the backward ratio recursion collapses to a spurious, problem-independent $R_n - 1 = 2/n$.

S2. BENCHMARKS

The five-dimensional Schwarzschild–Tangherlini fundamental scalar mode is reproduced to all 12 published digits,

$$\omega = 0.533835574268 - 0.383375368513 i \quad (r_+ = 1), \quad (\text{S1})$$

with absolute error 5.0×10^{-13} against the published value and 2.2×10^{-12} between Solvers A and B. Four static benchmarks ($\ell = 0, 1$; $n = 0, 1$) agree to 10^{-13} – 10^{-11} . Three two-spin points from the literature are reproduced, and the exchange symmetry $(a, m_1) \leftrightarrow (b, m_2)$ holds numerically to 4.2×10^{-15} . Three solvers agree below 10^{-5} on 10/10 benchmark points.

S3. ATLAS CENSUS

Quantity	Value
atlas points	50 558
independent sectors	7
angular branches per sector	3 ($\ell_{\min}, +2, +4$)
overtones	$N = 0, 1, 2, 3$
parameter points with ≥ 2 branches	4459
maximum continued-fraction residual	3.6×10^{-9}
predictor-miss events (recorded, not filled)	118
non-convergence events (recorded)	33
branch-label collapses excluded	4

TABLE SII. Branch atlas census. The atlas itself (≈ 24 MB) is regenerated by a deterministic driver and verified against a per-shard SHA-256 manifest rather than being distributed.

S4. DIAGNOSTICS AND THE VERIFICATION GATE

Each gate condition names the routine implementing it; each routine is validated on closed-form synthetic problems, including negative controls in which the detector must *not* fire.

Condition	Synthetic validation	Ten tightest candidates
G1 $ a_1/a_2 \rightarrow 0$ under refinement	recovers true separation to 10^{-6} ; invariant under $F \rightarrow 10^7 e^{3z} F$	fail
G2 argument-principle zero count = 2	counts zeros and poles; raises when unresolved	pass
G3 both roots located without a guess	finds both roots of a pair separated by 10^{-3}	pass
G4 monodromy transposes the branches	swaps for $\lambda^2 - t$; <i>does not</i> swap for $\lambda^2 - (t^2 + g^2)$	fail
G5 two loops restore the assignment	verified	—
G6 splitting $\sim t^{1/2}$, preferred over t^1	exponent 0.5 ± 10^{-6} ; rejects sqrt for linear splitting	fail
G7 Jordan chain solvable, geometric < algebraic	fires on a Jordan block; silent on semisimple and block-diagonal	—
G8 stable under resolution/depth/precision	solver level	—
G9 confirmed by a recurrence-free solver	cross-solver	—
G10 inside the validated solver domain	domain guard	pass

TABLE SIII. The EP2 gate. G2 and G3 are *preconditions*: they establish only that two distinct roots sit near each other. The decisive evidence is G1, G4 and G6, which fail uniformly, giving the classification “avoided crossing”.

An EP3 additionally requires three coalescing branches, zero count 3, $a_1 = a_2 = 0$ with $a_3 \neq 0$, cube-root splitting, a three-cycle monodromy with no fixed point, and Jordan-chain length 3.

S5. CERTIFICATION SCOPE

Certification claims must name the object certified. We use the labels: *arithmetic enclosed* (a ball enclosing the value of a given finite expression); *algebraic-system certified* (a Krawczyk certificate for a finite-dimensional nonlinear system); *truncated-recurrence certified* (the above with truncation error bounded); and *continuum certified* (a statement about the differential operator). Mandatory error separation:

$$E_{\text{total}} \leq E_{\text{discretization}} + E_{\text{truncation}} + E_{\text{arithmetic}} + E_{\text{root}}. \quad (\text{S2})$$

What is achieved: a Krawczyk existence/uniqueness test (validated in both directions — certifying a unique root, excluding a root-free box, and reporting “inconclusive” rather than guessing on an over-large box; it rejects a midpoint-only Jacobian, which is a heuristic rather than a certificate), and an Arb ball enclosure of the angular eigenvalue of the N -truncated Jacobi matrix, validated by recovering the exact value $\ell(\ell + 2)$ at $c^2 = 0$.

What is *not* achieved, and why: the radial problem is not certified. The polynomial coefficients of the recurrence are obtained by a discrete Fourier transform on a circle, and we have no rigorous bound on its aliasing error, so

$E_{\text{truncation}}$ cannot be bounded. This is the exact obstruction, and it is the reason no truncated-recurrence certificate is claimed anywhere. Closing it requires either a rigorous aliasing bound or an exact symbolic route to the coefficients that ball arithmetic can enclose end to end.

S6. FAILED APPROACHES, PRESERVED

Two branch-tracking defects each produced a small computed gap that looked physical. Both are recorded because the lesson generalizes: *a small computed gap between two labels is evidence about the labels before it is evidence about the physics.*

Branch-label collapse. With a coarse first step in s , the extrapolator has only one history point and degenerates to the trivial predictor. A strongly damped $\ell = 6$, $N = 3$ overtone was captured by the $N = 2$ branch while retaining its label, and the tier-1 search reported a minimum gap of 4.3×10^{-15} at 200 parameter points. It was caught by the rescaling-invariant diagnostic, which reported a separation of ≈ 1.0 at those same points — a flat contradiction, since a genuine coalescence requires both to be small. Fixed by finer early steps and a permanent collapse guard.

Branch jump admitted by a $|\omega|$ -relative tolerance. A tolerance $\propto |\omega|$ is a constant, and at $|\omega| \approx 3.7$ it admitted a jump in which $\text{Im}\omega$ reversed from -2.530 to -2.762 . The contaminated point was the reported tightest candidate. Fixed by taking the tolerance from the branch's own history of predictor misses, allowed $= \max(4 \text{ median}(\text{prior misses}), 0.02 \max(|\omega|, 1))$, and by reducing the s step to ≤ 0.05 ; an ≤ 0.05 and an ≤ 0.04 grid then give identical endpoints on all four hardest branches.

A withdrawn bound. An earlier exclusion in this project used $\min |dF/d\omega| = 1.02 \times 10^{-2}$. It is withdrawn: $|dF/d\omega|$ is not invariant under rescaling of the spectral condition. The conclusion survived re-derivation with invariant quantities; the reasoning did not.

A misdiagnosed regime. The failure of the recurrence-free solvers at large spin was attributed to near-extremality. It is not: ordinary modes continue to extremality 0.0103 with residuals $\sim 10^{-13}$. The controlling variable is μ , through $\text{Re}\Omega \rightarrow 0$.

S7. VERIFICATION OF THE RESIDUAL SYMMETRY

Figure S1 verifies Table I of the main text. In the diagonal sector $(1, 1)$ the relation $\omega(+\delta) = \omega(-\delta)$ holds to double precision; in the off-diagonal sector $(2, 0)$ it fails at the 10^{-2} level. The second curve is a negative control: it shows that the evenness is a property of the sector, not an artifact of a δ -blind solver.

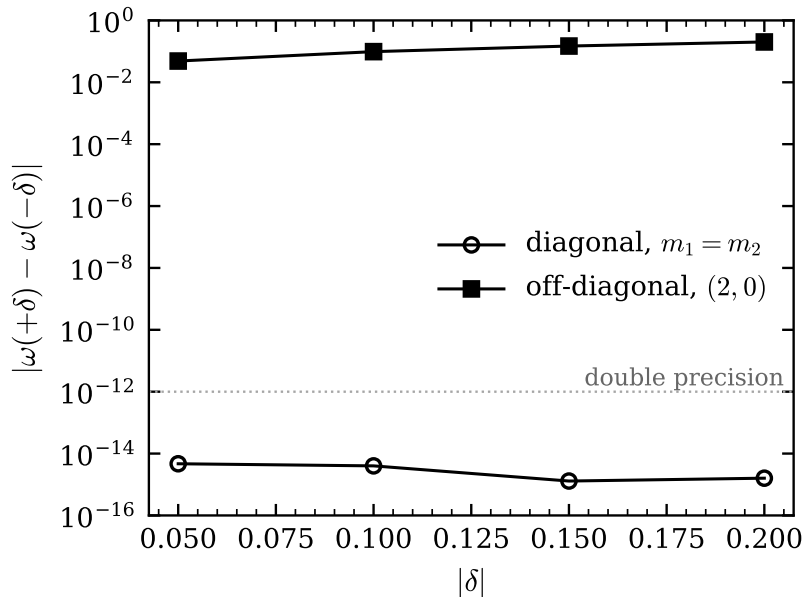


FIG. S1. Measured δ -parity, $\ell = 2$, $N = 0$, $s = 0.20$, $\mu = 0.6$.

S8. THE QUASIRESONANT REGIME

As the scalar mass grows, branches migrate toward the threshold $\omega^2 = \mu^2$ and become arbitrarily long lived. Tracking eleven branches across six sectors with an adaptive step in μ , the damping falls through 8.6 orders of magnitude along a single continuous branch, from 0.357 at $\mu = 0$ to 8.65×10^{-10} at $\mu = 2.968$, with $\text{Re } \omega \rightarrow \mu$ (Fig. S2). Each walk terminates where double precision can no longer resolve the mode, so the minima are lower bounds on how far a branch was followed, not infima. The transition is a branch point in μ , not a coalescence: the neighbouring overtone remains at $-\text{Im } \omega \sim 0.1$ throughout.

This regime also settles a question about solver behaviour that proves to be physics rather than numerics. Exterior complex scaling requires $\tan \vartheta > -\text{Im } \Omega / \text{Re } \Omega$. As $\text{Re } \Omega \rightarrow 0$ the outgoing wave ceases to oscillate and the required angle tends to 90° , so no admissible contour exists. We measure $\vartheta_{\min} = 89.92^\circ$ and 89.81° at the two points where the recurrence-free solvers fail, against 4.4° at an ordinary near-extremal point where they succeed. This is a limit of the formulation, not a convergence failure: no increase of resolution, precision or domain decomposition removes it, which is why a multidomain refinement of the near-horizon region did not help. It follows that the difficulty is controlled by μ , not by proximity to extremality — ordinary modes continue smoothly to extremality 0.0103 with residuals $\sim 10^{-13}$ over thirty-seven consecutive steps.

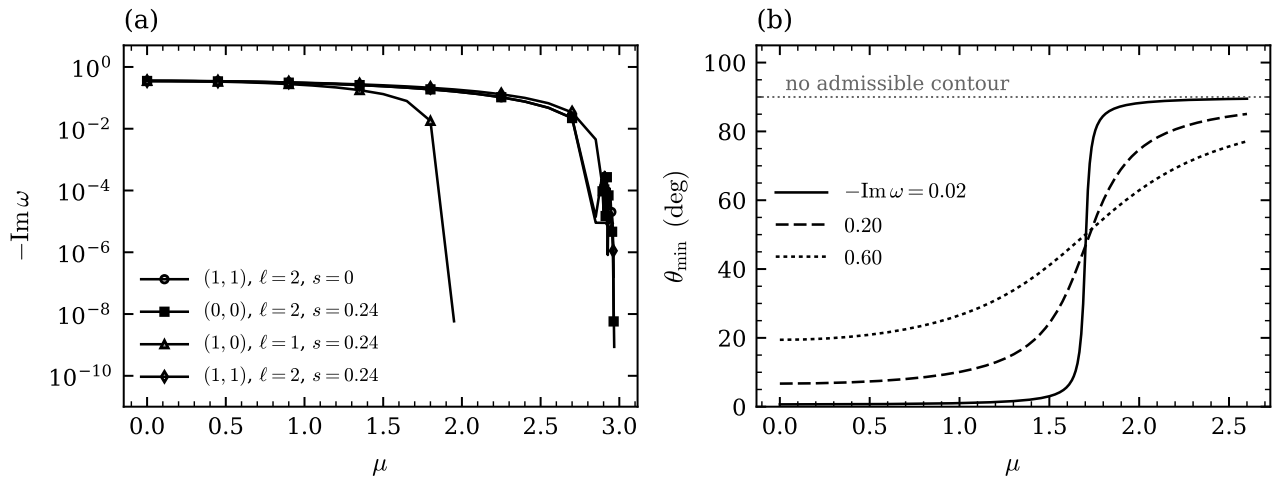


FIG. S2. (a) Damping along four branches continued in μ . (b) The minimum admissible exterior-complex-scaling angle; as $\text{Re } \Omega \rightarrow 0$ it reaches 90° and the method has no admissible contour.

S9. TWO DIAGNOSTICS WITHDRAWN

We record two diagnostics that were used in an earlier version of this work and are withdrawn, since both mistakes are natural and the second is subtle.

The ratio of leading Taylor coefficients. Under $F \mapsto gF$ with g holomorphic and nonvanishing, expanding about a simple root gives $\tilde{a}_1 = g_0 a_1$ and $\tilde{a}_2 = g_0 a_2 + g_1 a_1$, so $\tilde{a}_1 / \tilde{a}_2 = (a_1 / a_2) / [1 + (g_1 / g_0)(a_1 / a_2)]$: the ratio is only *asymptotically* insensitive to normalization, with distortion $\mathcal{O}(|g'|/g|d)$. Measured against $g = 10^7 e^{3\omega}$ the distortion follows $|1 + 3d|^{-1}$ to five decimals, from 1.0003 at $d = 10^{-4}$ to 1.4286 at $d = 0.1$. Decisively, the ratio measures the distance to the nearest zero *or pole*; Leaver's continued fraction is meromorphic, and contour integration about the tightest candidate returns $\mathcal{N} - \mathcal{P} = 0$ at every radius out to 0.20, so the small values returned are pole distances.

The discriminant as independent evidence. Since $D = (\omega_+ - \omega_-)^2$, its minimum is the square of the minimum separation — numerically $0.5116^2 = 0.26176$, agreeing to sixteen digits. It is one quantity reported twice, and is not counted separately.

A third defect found in the course of this: the diagnostics had been evaluated with the continued fraction inverted at index 0 irrespective of the branch's own overtone. The inversions are not equivalent at finite depth — at a tracked $N = 2$ root the $N = 2$ inversion gives $|F| = 1.9 \times 10^{-13}$ while inversions 0, 1, 3 give 7×10^{-2} , 1×10^{-1} and 1.1, values that are unchanged from depth 200 to depth 3200. A residual gate now rejects any expansion whose constant term is not small.

S10. REPRODUCTION

The scientific environment is locked; the plotting stack is deliberately kept out of it so that neither the results nor continuous integration depend on a graphics library. Verified from a fresh clone into an empty directory with a new virtual environment.

```
./scripts/reproduce_core_science.sh --quick    # analytical + machinery
./scripts/reproduce_core_science.sh           # + atlas + collision search
./scripts/reproduce_full_science.sh --resume   # everything
```